Minh Duc Hoang

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Summary

Second-year Data Science & AI student at HUST with a strong mathematical foundation and hands-on experience building LLM/NLP applications, from multi-agent systems to RAG pipelines. GDGoC Hackathon Vietnam 2026 champion and Head of the Developer Department at GDG-HUST. Aiming to grow as an AI Engineer, then pursue an MSc in AI/NLP.

Education

Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
Bachelor in Data Science and Artificial Intelligence — *Sep 2025 – Present*

- CPA: 3.88/4.0. Expected graduation: May 2029.
- Relevant Coursework: Calculus I&II, Algebra, Probability & Statistics, Discrete Math, DSA, Optimization.

Lam Son High School for the Gifted — Thanh Hoa, Vietnam
Specialization: English. GPA: 9.57/10 — *Sep 2022 – May 2025*

- National Salutatorian (A01 block), National High School Graduation Exam 2025: 29.5/30.0.

Experience

Google Developer Group on Campus - HUST (GDG-HUST) — Oct 2025 – Present
Head of Developer Dept. & Head of Data & AI (Aug 2026 – Present) — *Hanoi, Vietnam*

- Lead the Developer Department and set the technical direction for the club's Data & AI track.
- Plan activities, assign work, and provide technical support to members in projects and internal competitions.
- Earlier, as a Data & AI Developer, built housing price prediction and resume ranking models.

Projects & Competitions

Agicom – GDGoC Hackathon Vietnam 2026 | **Write-up** — Mar 2026 – May 2026
Team Leader, Backend & AI Engineer (team of 4) — *National Champion*

- Led team FUTURA to the national championship, outcompeting 330+ teams nationwide.
- Built Agicom, a multi-agent AI system acting as a 24/7 virtual operations team for e-commerce SMEs.
- Developed the multi-agent backend architecture using FastAPI.

Skillora – GDG-HUST Internal Hackathon 2026 — May 2026 – Jun 2026
Team Leader, Backend & AI Engineer (team of 4) — *Champion*

- Built a platform that uses an LLM to analyze students' GitHub repositories and map their skills into a skill tree.
- Built the FastAPI AI service and a Neo4j skill graph used for skill-gap analysis and learning recommendations.

Vietnamese Clinical Concept Extraction – Viettel AI Race 2026 — Jul 2026
Team Member (team of 2) — *Ranked 164/440 (Round 1)*

- Built a pipeline to extract medical concepts from Vietnamese clinical notes and map them to ICD-10/RxNorm.
- Compared rule-based, LLM-based and retrieval-based approaches using a local copy of the official metric.
- Experimented with LoRA fine-tuning of small open-source LLMs.

RAG for Multiple-choice QA – VNPT AI Hackathon: Age of AInicorns — Dec 2025
Team Leader, RAG pipeline (team of 4)

- Built a RAG pipeline that helps an LLM answer multi-domain multiple-choice questions.
- Handled document processing and hybrid retrieval (vector + BM25): 75.4% accuracy, 84.2% on STEM.

Skills & Qualifications

Programming: Python, C/C++, Java (OOP)
LLM & NLP: RAG, hybrid search, multi-agent systems, LoRA fine-tuning, Hugging Face, LangChain
ML & Data: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, OR-Tools, data preprocessing, error analysis
Backend & Databases: FastAPI, SQLAlchemy, REST APIs, PostgreSQL, SQLite, FAISS, ChromaDB
Tools & Deployment: Git, GitHub, Docker, Google Colab, Render, Netlify, VS Code, LaTeX
English: IELTS Academic 8.0 (2023); SAT 1580/1600 (2025)